

Assessing Risk: An Introduction

Module 3

Objectives

- Qualitative and quantitative risk analysis methods
- Analyse risk in terms of impact and likelihood of occurrence
- Use the standard risk model to score likelihood of risks and impacts
- Develop impact attributes to connect risk impacts



The Purpose of Risk Analysis

- Understand what is most uncertain about a risk
- Determine attention given to different risks
- Determine appropriate responses needed



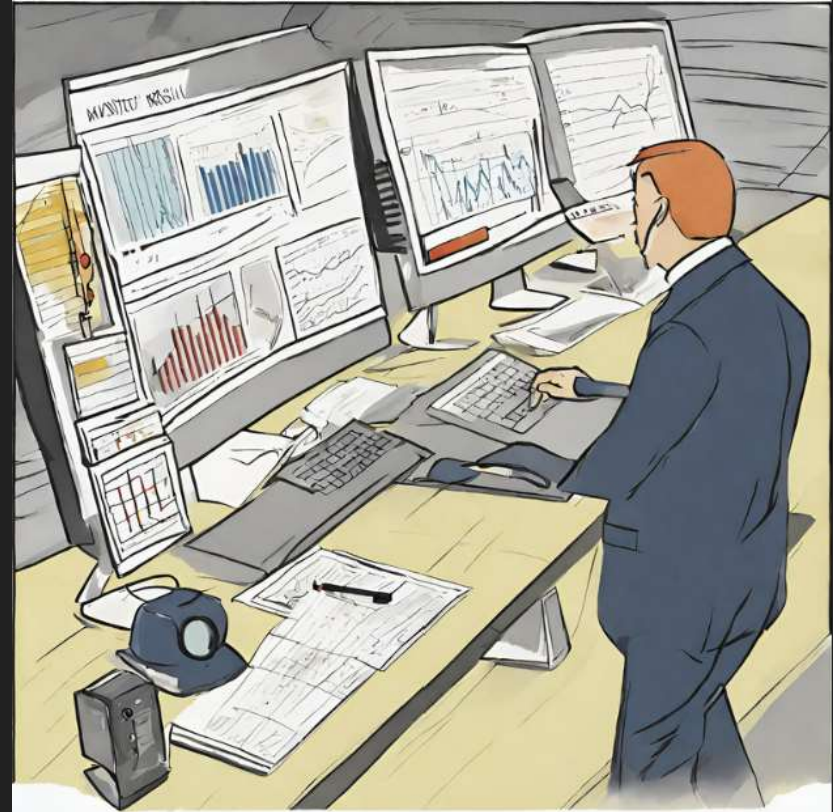
Qualitative and Quantitative methods

- Qualitative methods: to assess and prioritize risks - probability and impact matrix (risk heat map)
- Quantitative methods: to determine the magnitude of impact.



Risk analysis

- Understand risk probability (likelihood) and the risk impact
- Assure quality risk information
- Categorize the risks
- Identify the urgency of the risk (close to issue)



Three ways to assess risk probability

- Apply a probability of 1 to 9 to represent likelihood.
- Apply risk qualitative probabilities with a probability range of categories.
- Use a quantitative method which applies a specific value between 0 to 1 to each risk

Three Ways to Assess Risk Impact

- Qualitative- range or category of impact
- Quantitative- estimate of impact
- Non-measurable risk impact

Steps of Qualitative Risk Analysis

1. Select the appropriate qualitative risk analysis tool
2. Adapt tools to scale and scope of the organization
3. Apply tools to perform the qualitative risk analysis
4. Select risks for quantitative risk analysis
5. Update risk log and risk treatment and response plan

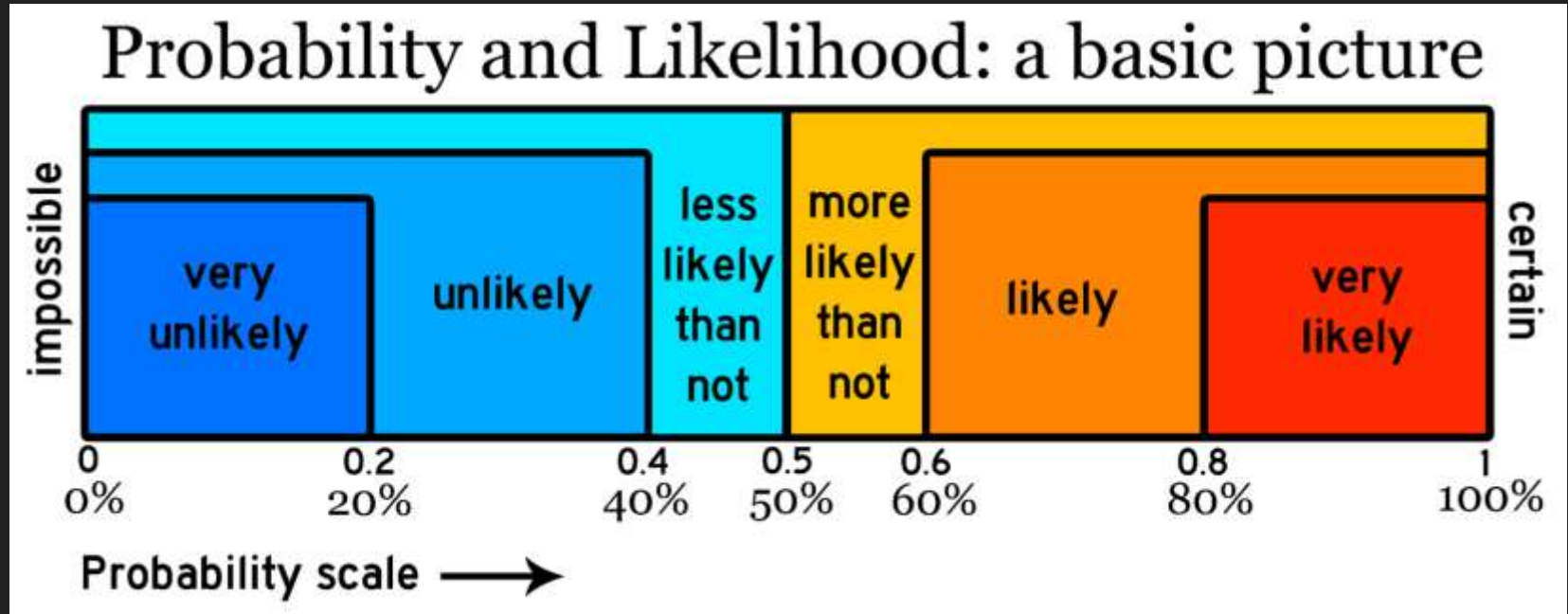
Primary Qualitative Risk Analysis Tools

- Risk Rating Schemes
- Probability/Impact Assessment & Matrix
- Urgency Assessment
- Risk Breakdown Structure
- Risk Factors

Secondary Qualitative Risk Analysis Tools

- Expert Judgement
- Planning Meetings
- Analogy Comparison
- Delphi Technique
- Root Cause Analysis
- Data Quality Assessment
- Risk Modeling

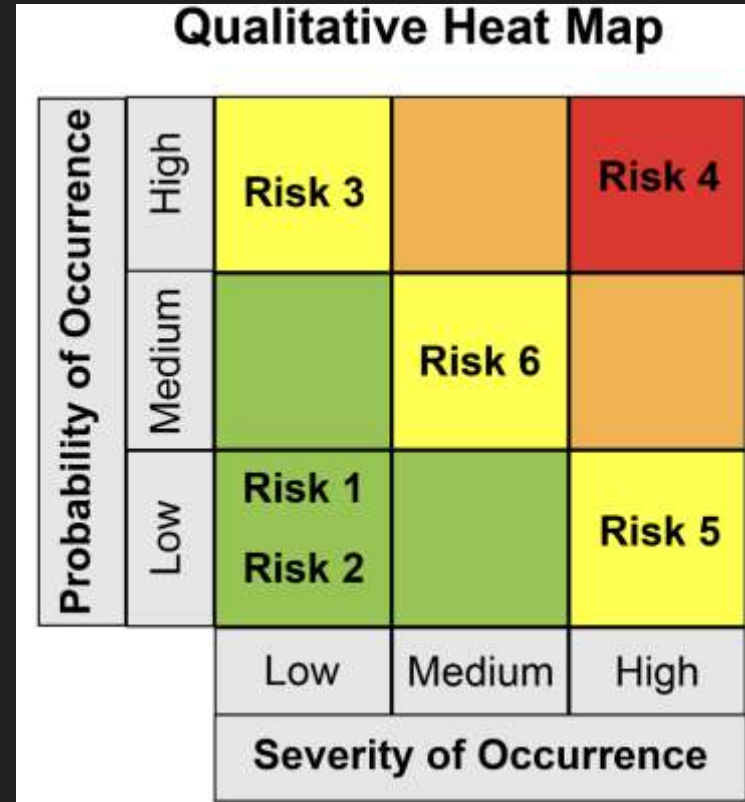
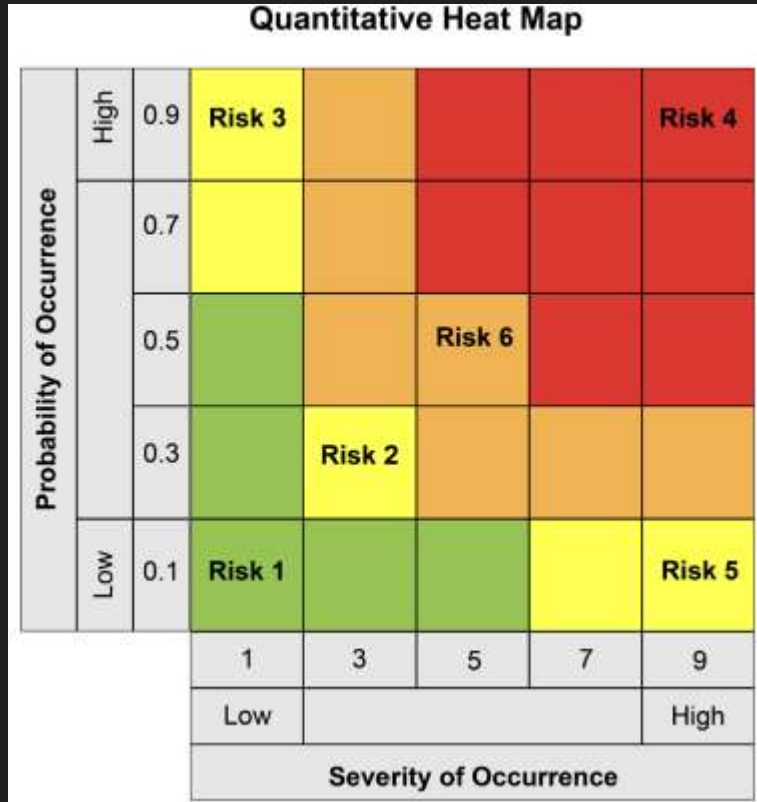
Risk Probability and Impact Assessment



Examples for scaling probabilities and impact

Probabilities (Likelihood)	Impact
Example 1: Quantitative	Example 1: Quantitative
Low: 0 to 15 percent	Low: 0 or 1
Medium: 15-50 percent	Medium: 3 or 5
High: 51 - 81 percent	High: 5 or 15
Example 2: Qualitative	Example 2: Qualitative
Rare	Very Low
Unlikely	Low
Moderate	Moderate
Likely	High
Highly Likely	Very High

Creating a Risk HeatMap



Identifying Urgency of Risk

- Time available to respond or mitigate the risk
- The likelihood of the risk occurring
- Impact of the risk to the organization
- If it becomes an issue
- Whether the organization can deal with it in a timely manner and pay for it.